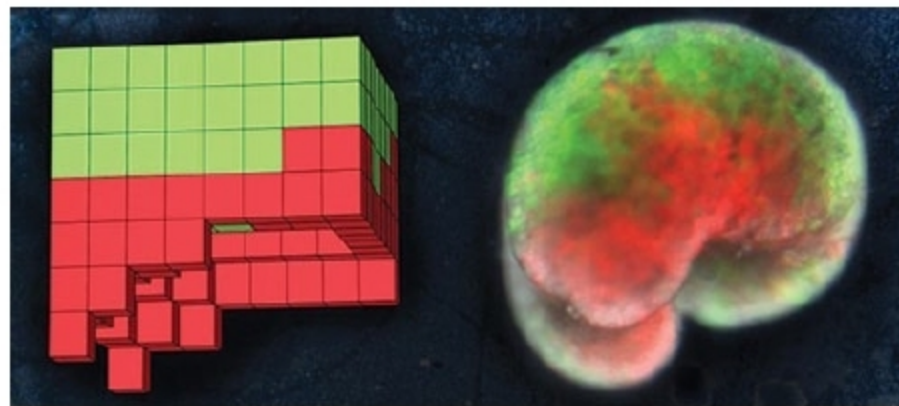


🔔 Researchers build the first living robots

A team of scientists has repurposed living cells—scraped from frog embryos—and assembled them into entirely new life-forms. These millimetre-wide xenobots can move towards a target, perhaps pick up a payload (like a medicine that needs to be carried to a specific place inside a patient) and heal themselves after being cut.

Joshua Bongard, computer scientist and robotics expert at University of Vermont (UVM), who co-led the new research, says, “These are novel living machines. They are neither a traditional robot nor a known species of animal. It is a new class of artefact—a living, programmable organism.”

The new creatures were designed on a supercomputer at UVM, and then assembled and tested by biologists at Tufts University. “We can imagine many useful applications of these living robots that other machines cannot do, like searching out nasty compounds or radioactive contamination, gathering microplastic in the oceans, travelling in arteries to scrape out



On the left, anatomical blueprint for a computer-designed organism, discovered on a UVM supercomputer. On the right, the living organism, built entirely from frog skin (green) and heart muscle (red) cells. The background displays traces carved by a swarm of these new-to-nature organisms as they move through a field of particulate matter (Credit: University of Vermont)

plaque,” says co-leader Michael Levin, who directs Center for Regenerative and Developmental Biology at Tufts.

🔔 Nano-thin flexible touchscreens could be printed like newspaper

Researchers have developed an ultra-thin and ultra-flexible electronic material that could be printed and rolled out like newspaper, for the touchscreens of the future. The touch-responsive technology is hundred times thinner than existing touchscreen materials and so pliable it can be rolled up like a tube.



A sample of the ultra-thin and ultra-flexible electronic material that could be printed and rolled out like newspaper, for the touchscreens of the future (Credit: RMIT University)

To create the new conductive sheet, the RMIT University-led team used a thin film common in cellphone touchscreens and shrunk it from 3D to 2D, using liquid metal chemistry.

The nano-thin sheets are readily compatible with existing electronic technologies and, because of their incredible flexibility, could potentially be manufactured through roll-to-roll (R2R) processing just like a newspaper.

The research was undertaken with collaborators from UNSW, Monash University and ARC Centre of Excellence in Future Low-Energy Electronics Technologies (FLEET).

Lead researcher, Dr Torben Daeneke, an Australian Research Council DECRA fellow at RMIT, says that most cellphone touchscreens are made of a transparent material, indium-tin oxide, which is very conductive but also very brittle. He adds, “We have taken an old material and transformed it from the inside to create a new version that is supremely thin and flexible.

“You can bend it; you can twist it and you could make it far more cheaply and efficiently than the slow and expensive way that we currently manufacture touchscreens. Turning it 2D also makes it more transparent, so it lets through more light. This means a cellphone with a touchscreen made of our material would use less power, extending the battery life by roughly ten per cent.”

🔔 Researchers reveal new Chameleon attack that can secretly modify content on Facebook, Twitter or LinkedIn

That video or picture you liked on social media of a cute dog, your favourite team or political candidate can be altered in a cyberattack to something completely different, detrimental and potentially criminal, according to cybersecurity researchers at Ben-Gurion University of the Negev (BGU).

BGU researchers looked at seven online platforms and identified similar serious weaknesses in the management of the posting systems of Facebook, Twitter and LinkedIn. Twitter does not permit changes to posts and, normally, Facebook and LinkedIn indicate a post has been edited. But this new